

Def. Let V be a real or Complex vector space. A function $f: V \times V \rightarrow F$ such that
 for all $\alpha, \beta, \gamma \in V$ and $c \in F$, $\begin{cases} f(c\alpha + \beta, \gamma) = c \cdot f(\alpha, \gamma) + f(\beta, \gamma) \\ f(\alpha, c\beta + \gamma) = \bar{c} \cdot f(\alpha, \beta) + f(\alpha, \gamma) \end{cases}$
 is called the Sesqui-linear form.

Thm. Let V be a finite-dimensional inner product space, and f be a sesqui-linear form on V .
 Then, there is a unique linear operator T on V such that

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle \text{ for all } \alpha, \beta \in V.$$

Moreover, a map $f \mapsto T$ is an Isomorphism.

Proof. Given $\beta \in V$, $\alpha \mapsto f(\alpha, \beta)$ is a linear functional on V .

By the Riesz Representation Theorem, there exists a unique vector $\beta' \in V$ such that

$$f(\alpha, \beta) = \langle \alpha, \beta' \rangle \text{ for all } \alpha \in V.$$

Now, let $U: V \rightarrow V$ be an mapping β to β' . Then, given $\alpha, \beta, \gamma \in V$ and $c \in F$,

$$\begin{aligned} f(\alpha, c\beta + \gamma) &= \langle \alpha, U(c\beta + \gamma) \rangle \text{ and} \\ &= \bar{c} \cdot f(\alpha, \beta) + f(\alpha, \gamma) = \bar{c} \langle \alpha, U(\beta) \rangle + \langle \alpha, U(\gamma) \rangle = \langle \alpha, c \cdot U(\beta) + U(\gamma) \rangle \end{aligned}$$

Since α was arbitrary, U is linear. Since V is finite, adjoint of U exists.
 Put $T = U^*$. Then, the T is we desired. The uniqueness is clear.

In sum, for each form f , there is a unique linear operator T_f such that

$$f(\alpha, \beta) = \langle T_f(\alpha), \beta \rangle \quad \text{Existence and Uniqueness.}$$

Given forms f and g and scalar c ,

$$\begin{aligned} (cf + g)(\alpha, \beta) &= \langle T_{cf+g}(\alpha), \beta \rangle \text{ and} \\ &= cf(\alpha, \beta) + g(\alpha, \beta) = \langle (cT_f + T_g)(\alpha), \beta \rangle \end{aligned}$$

So $T_{cf+g} = cT_f + T_g \downarrow f \mapsto T_f$ is linear

And surjective is clear, and $T_f = 0 \Leftrightarrow f = 0$, so injective. $\square$

Corollary. Given forms $f, g: V \times V \rightarrow F$, define

$$\langle f, g \rangle \stackrel{\text{def}}{=} \text{Tr}(T_f T_g^*) \stackrel{(*)}{=} \sum_{j,k} f(\alpha_k, \alpha_j) \cdot \overline{g(\alpha_k, \alpha_j)}$$

is an inner product on $V \times V$ such that $(*)$, for any orthonormal basis $\{\alpha_1, \dots, \alpha_n\}$ for V .

Proof. Given orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$, let $A = [T_f]_{\mathcal{B}}$ and $B = [T_g]_{\mathcal{B}}$.

Then, by the orthonormality and the above Thm,

$$A_{jk} = \langle T_f(\alpha_k), \alpha_j \rangle = f(\alpha_k, \alpha_j) \text{ and } B_{jk} = \langle T_g(\alpha_k), \alpha_j \rangle = g(\alpha_k, \alpha_j).$$

Moreover, $[T_f \circ T_g^*]_{\mathcal{B}} = [T_f]_{\mathcal{B}} \cdot [T_g^*]_{\mathcal{B}} = AB^*$, so $\text{Tr}(AB^*) = \sum_{j,k} A_{jk} \cdot \overline{B_{jk}}$ gives $(*)$.

Def. Let V be a fin. dim inner product space, with basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$.

For a given form $f: V \times V \rightarrow F$, the matrix with entries

$$A_{jk} = f(\alpha_k, \alpha_j)$$

is called the matrix of f in the ordered basis $\mathcal{B}$.

Observe that: for a form $f: V \times V \rightarrow F$ and orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$.

Then, $f\left(\sum_s \chi_s \cdot \alpha_s, \sum_r \gamma_r \cdot \alpha_r\right) = \sum_{r,s} \overline{\gamma_r} A_{rs} \chi_s$, or

$$f(\alpha, \beta) = Y^* A X \text{ where } X = [\alpha]_{\mathcal{B}}, Y = [\beta]_{\mathcal{B}}.$$

Thm. Let V be a fin. dim Complex inn. Prod space.

For any form on $V \times V$, there is an orthonormal basis for V , in which the matrix of f is upper-triangular.

Proof. Since V is fin. dim. Complex. inn. Prod. space, for linear operator $T: V \rightarrow V$ s.t.

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle,$$

it has an orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ for V s.t. $[T]_{\mathcal{B}}$ is upper-triangular. So,

$$f(\alpha_k, \alpha_j) = \langle T(\alpha_k), \alpha_j \rangle = 0 \text{ if } j > k.$$

Def. A form f on V is called Hermitian if

$$f(\alpha, \beta) = \overline{f(\beta, \alpha)} \quad \text{for all } \alpha, \beta \in V.$$

Note: if there is a linear operator T on V such that $f(\alpha, \beta) = \langle T(\alpha), \beta \rangle$, then $\overline{f(\beta, \alpha)} = \langle \alpha, T(\beta) \rangle = \langle T^*(\alpha), \beta \rangle$, so,

f is Hermitian if and only if T is Hermitian.

If T is Hermitian, then $\langle T(\alpha), \alpha \rangle$ is real for all $\alpha \in V$, so $f(\alpha, \alpha)$ is real.

Lemma. Let V be a Complex inner product space, and f is a form on V such that $f(\alpha, \alpha)$ is real, for all $\alpha \in V$.

Then, f is Hermitian.

Proof. Given $\alpha, \beta \in V$,

$$f(\alpha + \beta, \alpha + \beta) = f(\alpha, \alpha) + f(\beta, \beta) + f(\alpha, \beta) + f(\beta, \alpha)$$

$$f(\alpha + i\beta, \alpha + i\beta) = f(\alpha, \alpha) + f(i\beta, i\beta) - i \cdot f(\alpha, \beta) + i f(\beta, \alpha)$$

By the assumption, $f(\alpha, \beta) + f(\beta, \alpha)$ and $-i \cdot f(\alpha, \beta) + i \cdot f(\beta, \alpha)$ are real. Thus,

$$f(\alpha, \beta) + f(\beta, \alpha) = \overline{f(\alpha, \beta)} + \overline{f(\beta, \alpha)} \quad \text{and} \quad -i \cdot f(\alpha, \beta) + i \cdot f(\beta, \alpha) = i \cdot \overline{f(\alpha, \beta)} - i \cdot \overline{f(\beta, \alpha)}$$

Multiplying i to the right, then we obtain

$$f(\alpha, \beta) - f(\beta, \alpha) = -\overline{f(\alpha, \beta)} + \overline{f(\beta, \alpha)}$$

so, $2f(\alpha, \beta) = 2 \cdot \overline{f(\alpha, \beta)}$, or f is Hermitian. $\square$

Combining the above note with this Lemma if V is fin. dim, then

T is Hermitian if and only if $\langle T(\alpha), \alpha \rangle$ is real.

Principal Axis Theorem.

Thm. For every Hermitian form f on a fin. dim inn. prod space V , there exists an orthonormal basis $\beta = \{\alpha_1, \dots, \alpha_n\}$ such that

f is represented by a diagonal matrix with real entries.

Proof. Since V is fin. dim, there exists a linear operator T on V s.t.

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle.$$

And, by the lemma above, f is Hermitian iff T is Hermitian, so there is an orthonormal basis for V which is consisted by eigenvector of T . Denote $\beta = \{\alpha_1, \dots, \alpha_n\}$, and C_i be the eigenvalue corresponding to α_i . Then,

$$f(\alpha_k, \alpha_i) = \langle T(\alpha_k), \alpha_i \rangle = C_k \cdot \delta_{ki},$$

this proved. (C_k are real by T is Hermitian). ~~Q~~

Note: Under the condition hold,

$$f\left(\sum x_i \alpha_i, \sum y_k \alpha_k\right) = \sum_i C_i x_i \overline{y_i}$$